

Practice Worksheet 5: Iteration II

CS 101 Algorithmic Problem Solving

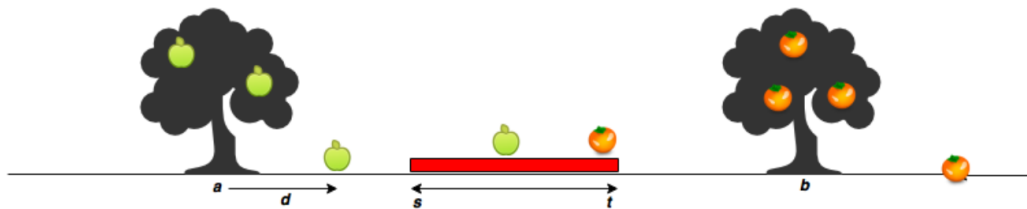
Fall 2024

Name(s): Syed Sualeh Abbas Rizvi

HU ID (e.g., xy01042): 10303

1. Fallen Fruits [1 Point]

Mohammad's house has an apple tree and an orange tree that yield an abundance of fruit. Using the information given below, determine the number of apples and oranges that land on Mohammad's house. (refer to the image provided)



Important Points To Consider

- The red shaded region denotes the house, where s is the start point, and t is the endpoint. The apple tree is to the left of the house, and the orange tree is to its right.
- Assume the trees are located on a single point, where the apple tree is at point a , and the orange tree is at point b .
- When a fruit falls from its tree, it lands d units of distance from its tree of origin along the x -axis. A negative value of d means the fruit fell d units to the tree's left, and a positive value of d means it falls d units to the tree's right.

Given the value of d for m apples and n oranges, determine how many apples and oranges will fall on Sam's house (i.e., in the inclusive range $[s, t]$)?

Constraints

- $1 \leq s, t, a, b, m, n \leq 1000$
- $-1000 \leq d \leq 1000$
- $a \leq s \leq t \leq b$

Interaction

The input comprises a single line containing 6 space-separated integers denoting the values of s, t, a, b, m and n respectively. Input has two more lines. The second line contains m space-separated integers denoting the respective distances that each apple falls from point

a. The third line contains n space-separated integers denoting the respective distances that each orange falls from point b .

The output must print two space-separated integers. First integer representing the number of apples and second integer representing the number of oranges.

Sample

Input	Output
7 11 5 15 3 2	1 1
-2 2 1	
5 -6	

In the above case: the first apple falls at position $(5 - 2) = 3$, second falls at $(5 + 2) = 7$, and the third falls $(5 + 1) = 6$.

The first orange falls at $(15 + 5) = 20$ and the second falls at $(15 - 6) = 9$.

One apple and one orange falls within the range of 7 and 11 hence, the output is 1 and 1.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
8 13 6 18 4 3	2 1
3 -2 2 1	
-4 -6 2	

Pseudocode

Input: s,t,a,b,m,n

Output:

orange=0

for i in range(m):

 m1=int(input("enter distance of apple here: "))

 if a+(m1) in range(s,t+1):

 apple+=1

 else:

 apple+=0

for l in range(n):

 n1=int(input("enter distance of orange here "))

 if b+(n1) in range(s-1,t+1):

 orange+=1

 else:

 orange+=0

print(apple, orange)

Dry Run

Below, dry run your pseudocode on any **one** of the inputs provided in the Exercise section.

s	t	a	b	m	n
8	13	6	18	4	3
				m1=3 (true)	n1=-4 (false)
				m2=-2 (false)	n2=-6 (true)(orange=1)
				m3=2 (true)	n3=2 (false)
				m4=1(false)	
				total apples = 2	total orange = 1

2. Good Boy [1 Point]

You adopted a dog not too long ago that has a knack for solving mathematical problems. You write things on a white board and the dog responds by pressing keys on a large keypad you built for him.

Recently, you've discovered that he's quite good at performing the $sum(N)$ function, i.e. summing all natural numbers up until the specified number N . He's gotten so good in fact, that he can easily perform this function multiple times in one go, giving you the answer faster than it would take you to type in a calculator.

Your dog has expertly defined this repeated sum function as $sum(D, N)$, which denotes the $sum(N)$ function applied D times repeatedly: the first time it sums up till N , and each subsequent time to the result of the previous sum function. For example, $sum(2, 3)$ is equivalent to $sum(sum(3)) = sum(1 + 2 + 3) = sum(6) = 21$.

You have found this repeated sum function to be quite useful, and want to try your own hand at it for times when your dog is not there to help you out.

Given values of D and N , determine the output of the $sum(D, N)$ function.

Constraints

- $D, N \in \mathbb{N}$
- $1 \leq D, N \leq 10^5$

Interaction

The input comprises of a single line containing 2 space-separated integers denoting the values of D and N respectively.

The output must be a single integer denoting the value of $sum(D, N)$.

Sample

Input	Output
5 2	26796
1 10	55

In the first case, $(D, N) = (5, 2)$. $sum(5, 2)$ is equivalent to $sum(sum(sum(sum(sum(2)))))) = 26796$.

In the second case, $(D, N) = (1, 10)$. $sum(1, 10)$ is equivalent to $sum(10) = 55$.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
2 6	231
112 1	1
3 2	21

Pseudocode

Input: D, N

Output:

```
D=int(input("enter value of D: "))
```

```
N=int(input("enter value of N: "))
```

```
for i in range(D):
```

```
    N = N * (N + 1) / 2
```

```
print(int(N))
```

Dry Run

Below, dry run your pseudocode on any **one** of the inputs provided in the Exercise section.

Input:

D=112

N=1

```
for i in range(112): (i=1)
```

```
    N = 1 * (1 + 1) / 2 (N=1.0)
```

```
for i in range(112): (i=2)
```

```
    N = 1.0 * (1.0+1) / 2 (N=1.0)
```

```
for i in range(112): (i=3)
```

```
    N = 1.0 * (1.0+1) / 2 (N=1.0)
```

and like this the loop will run 112 times, as the value of N remains constant, it will print the value after the code ends. (N=1.0)

```
print(int(1.0))
```

Output: 1

3. Lucky Days [1 Point]

You and a group of friends have been obsessed with the idea of “lucky days”. A lucky day is one where, if the date is represented in the format DD/YY/MM, the date is palindromic OR if changing any ONE digit in the date’s representation makes it palindromic. Your friends have compiled a dataset of 6 digit long integers representing dates from the 20th century, such that the first 2 digits represent the day, the middle 2 represent the year, and the last two represent the month. For example, 30th March 2022 would be represented as 302203. Given an integer N , determine if it represents a lucky day.

Note: You are not allowed to convert the integer N into a string.

Constraints

- $N \in \mathbb{N}$
- N contains 6 digits long. The first digit can be 0.
- The first 2 digits of N are between 01 and 31, the middle 2 digits are between 00 and 99, and the last 2 digits are between 01 and 12.

Interaction

The input comprises of a single line containing a single 6 digit long integer denoting the value of N .

The output must be “YES” if the integer represents a lucky day, and “NO” otherwise.

Sample

Input	Output
213212	“YES”
218812	“YES”

In the first case, $N = 213212$. Changing the 4th digit in N to a 3 makes it 213312 which is palindromic and hence, it is a lucky day.

In the second case, $N = 218812$. As N backwards is $218812 = N$, it is a lucky day.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
204412	YES
121212	NO
101011	NO

Pseudocode

```

Input: N
e=N//1000
b = 0
N1 = N
f=0
a=0
l = True

while N > e:
    c = N % 10
    N = N // 10
    d = b % 10
    b = b // 10
    print(c,d)
    if abs(c-d)==0:
        l = True
    elif abs(c - d) == 1:
        a=a+1
    else:
        l=False
        f=f+1
    if l == False or f>0:
        print("NO")
    elif l==True and a<=1:
        print("YES")
    else:
        print("NO")

for i in range(6):
    b = (b * 10) + N1 % 10
    N1 = N1 // 10
  
```

Dry Run

Below, dry run your pseudocode on any **one** of the inputs provided in the Exercise section.

<pre> N=121212 e=N//1000 (e=121) b = 0 N1 = N f=0 a=0 l = True for i in range(6): b = (b * 10) + N1 % 10 (b=2) N1 = N1 //10 (121212) </pre>	<pre> while 121212 > 121: c = N % 10 (c=2) N = N // 10 (N=12121) d = b % 10 (d=1) b = b //10 if abs(c-d)==0: (false) l = True elif abs(c - d) == 1: (true) a=a+1 (a=1) while 12121 > 121: c = N % 10 (c=1) N = N // 10 (N=1212) d = b % 10 (d=2) b = b //10 if abs(c-d)==0: (false) l = True elif abs(c - d) == 1: (true) a=a+1 (a=2) </pre>	<pre> while 1212 > 121: c = N % 10 (c=2) N = N // 10 (N=121) d = b % 10 (d=1) b = b //10 if abs(c-d)==0: (false) l = True elif abs(c - d) == 1: (true) a=a+1 (a=3) while 121 > 121: (false) if l == False or f>0: (false) print("NO") elif l==True and a<=1: (false) print("YES") else: (true) print("NO") Output= "NO" </pre>
--	---	---

4. Unique Numbers [1 Point]

You are given two baskets, X_1 and X_2 . X_1 contains all integers in the range $[A, B]$ and X_2 contains all integers in the range $[C, D]$. You want to combine these two baskets into a third basket such that there are no duplicate elements.

Given the values of A, B, C , and D , determine the size of the third basket.

Constraints

- $A, B, C, D \in \mathbb{N}$
- $1 \leq A \leq B \leq 20$
- $1 \leq C \leq D \leq 20$

Interaction

The input comprises a single line containing 4 space-separated integers denoting the values of A, B, C , and D respectively.

The output must contain a single number denoting the size of the third basket.

Sample

Input	Output
5 9 7 11	7
5 9 11 15	10

In the first case, $(A, B, C, D) = (5, 9, 7, 11)$. There are 5 integers in X_1 and 5 integers in X_2 , but there's 3 integers i.e. 7, 8, 9 which are in both X_1 and X_2 , so the size of the third basket is 7.

In the second case, $(A, B, C, D) = (5, 9, 11, 15)$. There are 5 integers in X_1 and 5 integers in X_2 , and no such integers which are in both X_1 and X_2 , so the size of the third basket is 10.

Proposed Solution

```
count = (B-A) + 1 # counting all the numbers in [A,B]
for i in range(C, D):
    if i > B or i < A:
        count += 1
print(count)
```

Dry Run

Below, dry run the proposed pseudocode on the inputs provided in the Sample section.

Input: 5, 9,11,15 (A=5, B=9, C=11, D=15)

count = 5

for i in range (11,15):

i	condition	count
11	true	6
12	true	7
13	true	8
14	true	9

Output: 9

Error Identification

Briefly explain the errors you identified in the proposed code solution. Mention the line number and the errors in each line.

The range of for loop in line 2 actually means the code will run D-1 times. The range should be (C , D+1) instead of (C, D).

Correct Solution

Rewrite the lines of code you mentioned above with their errors corrected.

```
count = (B-A) + 1
for i in range (C , D+1):
    if i > B or i < A:
        count += 1
print ( count )
```

5. Intoduction to Hackerrank [6 Points]

The guide to login/sign-up on Hackerrank tests is available here.

Problems are available on this link as a short introduction to programming:

<https://hr.gs/cs101-pw5-fall24>

Rough Work